

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY
Bachelor of Engineering - SEMESTER - VI EXAMINATION - WINTER 2025

Subject Code: 3160716

Date: 19-11-2025

Subject Name: IOT and applications

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

- 1. Attempt all questions.**
- 2. Make suitable assumptions wherever necessary.**
- 3. Figures to the right indicate full marks.**
- 4. Simple and non-programmable scientific calculators are allowed.**

	Marks
Q.1 (a) Explain the role of "things" in IoT	03
(b) Explain the significance of AI/ML in IoT.	04
(c) With neat diagrammatic representation, explain IoT Stack with appropriate examples for each layer.	07
Q.2 (a) What are the different categories of the Sensors?	03
(b) Mention clearly the procedure to interface the Ultrasonic sensor with any microcontroller of your choice.	04
(c) What are the advantages of IoT in Health care sector? Discuss the role of IoT in Health Care Sector with proper example.	07
OR	
(c) What are the advantages of IoT in Agriculture sector? Discuss the role of IoT in Agriculture Sector with proper example.	07
Q.3 (a) Write short note on: Zigbee.	03
(b) Give the differences between M2M and IoT.	04
(c) Draw the BLE stack and explain its components.	07
OR	
(a) Explain the importance of URI in networking.	03
(b) How is IPv4 inferior to IPv6?	04
(c) Explain the importance of MQTT protocol with respect to IoT infrastructure	07
Q.4 (a) List the various Cloud deployment Models.	03
(b) What is fog computing? List the characteristics of fog computing.	04
(c) Explain case study of Adafruit cloud.	07
OR	
(a) What are the common threats to cloud storage?	03

- (b) What are the major security challenges in IoT compared to traditional networks? 04
- (c) Describe lightweight cryptography and why it is important for IoT devices? 07
- Q.5** (a) How can IoT improve warehouse management? 03
- (b) What is Arduino? Explain features of Arduino architecture. 04
- (c) Write arduino code to realize Traffic light controller using RED, GREEN & ORANGE LED. At the interval of 30 second, traffic lights should be turned on-off in a specific sequence. 07

OR

- (a) What type of IoT devices are commonly used in driver assistance systems? 03
- (b) Draw the block diagram of Raspberry Pi and explain components of Raspberry Pi in brief. 04
- (c) Write arduino code to interface Temperature sensor & LED with Arduino UNO board. If the temperature value is greater than 35 degree Celsius, LED should be turn on. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI EXAMINATION – SUMMER 2025****Subject Code: 3160716****Date: 26-05-2025****Subject Name: IOT and applications****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1*	(a) Describe Internet of Things.	03
	(b) (i) List the sources of IoT development board which can be used for prototype development. (ii) List communication protocols used in IoT.	04
	(c) Explain various levels of IoT.	07
Q.2	(a) List the sensors that may be useful for Smart Home.	03
	(b) What do you mean by actuators? Give examples of commonly used actuators in IoT.	04
	(c) Explain the key features of the ARM Cortex-M architecture that make it suitable for IoT.	07
	OR	
	(c) Define following properties of sensors: i. Accuracy ii. Precision iii. Drift iv. Resolution	07
Q.3	(a) What do you understand by 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks)?	03
	(b) How IPv6 address limitations of IPv4 in context of IoT?	04
	(c) In context of MQTT (Message Queuing Telemetry Transport) answer following questions in brief: (i) Define broker in MQTT? (ii) In MQTT, what is the communication pattern(architecture) used to send and receive messages? (ii) In MQTT, which QoS level ensures that a message is delivered exactly once without duplication? (iv) Give the use of '#' wildcard for topic subscription.	07
	OR	
Q.3	(a) Compare CoAP's request/response model with HTTP's request/response model. How does CoAP's asynchronous nature make it more suitable for constrained networks?	03
	(b) Describe the four main types of messages used in CoAP (Confirmable, Non-confirmable, Acknowledgment, and Reset). How are these messages used to ensure reliable communication?	04
	(c) Discuss the importance of cloud security in IoT.	07
Q.4	(a) Explain Bluetooth Low Energy (BLE).	03

	(b)	How does cloud computing enhance the scalability, storage, and computational capabilities of IoT systems?	04
	(c)	What do you mean by Fog computing? Discuss the advantages and disadvantages of fog computing for IoT applications compared to pure cloud solutions.	07
	OR		
Q.4	(a)	List sensors which can be used for Driver Assistance.	03
	(b)	How is IoT transforming patient care in healthcare? Discuss the role of wearable devices and remote monitoring systems in improving patient outcomes and reducing hospital visits.	04
	(c)	What are various cloud service models (IaaS, PaaS, SaaS). Discuss in detail in context of IoT application development.	07
Q.5	(a)	Compare Raspberry pi with Arduino.	03
	(b)	How many digital and analog input pins are available on the Arduino Uno? What is the difference between them?	04
	(c)	Write the purpose of following functions in Arduino: i. loop() ii. setup() iii. delay() iv. pinMode()	07
	OR		
Q.5	(a)	List enabling technologies for IoT.	03
	(b)	Explain the potential privacy concerns associated with IoT devices in smart homes and healthcare systems.	04
	(c)	Discuss various security issues in IoT?	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-VI (NEW) EXAMINATION – WINTER 2024****Subject Code:3160716****Date:28-11-2024****Subject Name:IOT and applications****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) What is IoT ? List any five IoT applications.	03
	(b) Explain enabling technologies in IoT.	04
	(c) Draw and explain IOT architecture.	07
Q.2	(a) Draw and explain IPv6 Header	03
	(b) Explain various levels of IoT.	04
	(c) List out types of IoT protocols and explain any one in brief.	07
	OR	
	(c) Explain pinMode(), digitalRead() and digitalWrite () functions of Arduino.	07
Q.3	(a) Give difference between IPV4 and IPV6 transport protocol.	03
	(b) What is Sensor ? List different types of Sensors used to develop IOT.	04
	(c) Explain Security and Privacy in cloud with respect to IoT based application.	07
	OR	
Q.3	(a) Explain CoAP. List the key features of CoAP.	03
	(b) Explain limitation of cloud computing.	04
	(c) List the characteristics of fog computing. Explain fog computing in detail.	07
Q.4	(a) Explain MQTT. List the key features of MQTT.	03
	(b) Write Raspberry Pi code to change the brightness of LED.	04
	(c) What is Arduino? Explain features of Arduino architecture.	07
	OR	
Q.4	(a) What is PWM? Explain usage of PWM pins with Example.	03
	(b) Write Raspberry Pi code to blink LED ON and OFF.	04
	(c) Explain interfacing of LED with Raspberry Pi.	07
Q.5	(a) Explain various application of IoT in Smart City.	03
	(b) Explain IoT security challenges.	04
	(c) Draw and explain architecture of IoT security.	07
	OR	
Q.5	(a) Explain various application of IoT in Smart Home.	03
	(b) Explain any one algorithm for IoT security.	04
	(c) What are the security and privacy issues of IOT? Explain using suitable case scenario.	07

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2024

Subject Code:3160716

Date:20-05-2024

Subject Name:IOT and applications

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

		MARKS
Q.1	(a) Explain Sensor interfacing.	03
	(b) Explain the Characteristics of IoT.	04
	(c) Explain challenges of IoT.	07
Q.2	(a) What is IP addressing?	03
	(b) Compare IPv4 and IPv6 in brief.	04
	(c) Explain various levels of IoT.	07
	OR	
	(c) List security issues and needs in IoT. Justify the answer.	07
Q.3	(a) Explain IoT and cyber physical system.	03
	(b) Explain CoAP. List the key features of CoAP.	04
	(c) Explain with diagram IPv6 header format	07
	OR	
Q.3	(a) List and explain cloud components.	03
	(b) Explain the brain in an IoT device.	04
	(c) Explain with diagram IPv4 header format	07
Q.4	(a) What is Messaging protocols.	03
	(b) Discuss Security in cloud.	04
	(c) Compare IoT and cloud.	07
	OR	
Q.4	(a) Explain Warehouse application with IoT.	03
	(b) Explain CSPR in ARM.	04
	(c) Discuss various application of IoT.	07
Q.5	(a) Explain Arduino in brief.	03
	(b) What does the future hold for smart home technology?	04
	(c) Draw and explain IoT based collision impact measuring system architecture.	07
	OR	
Q.5	(a) Explain in brief future factory concepts	03
	(b) Explain Raspberry Pi.	04
	(c) What risks do insecure IoT devices bring to privacy and security?	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – WINTER 2023****Subject Code:3160716****Date:07-12-2023****Subject Name:IOT and applications****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

Q.1	(a) Define IOT. Also Write the merits and demerits of IOT.	03
	(b) Discuss smart health using IOT.	04
	(c) Explain the architecture of the IOT.	07
Q.2	(a) Which are the network and communication technologies can be used to implement an IOT application?	03
	(b) Explain IOT value chains.	04
	(c) For building an architecture of IOT explain main design principles and needed capabilities.	07
OR		
	(c) Which are hardware and software components of IOT? How do these components get synchronised in an IOT based application?	07
Q.3	(a) Explain IOT Domain Model.	03
	(b) What is cloud? What is the role of cloud platforms in IOT?	04
	(c) What is ARM? Explain special feature of ARM processor.	07
OR		
Q.3	(a) Differentiate between IPv4 and IPv6.	03
	(b) What are the security challenges of IOT?	04
	(c) List and explain cloud deployment models.	07
Q.4	(a) Differentiate between M2M and IOT.	03
	(b) Explain IPv4 header format with diagram.	04
	(c) Explain case study of Adafruit cloud.	07
OR		
Q.4	(a) What is fog computing? List the characteristics of fog computing.	03
	(b) Discuss various register set of 8051.	04
	(c) What is cloud analytics? Explain in brief future factory concepts.	07
Q.5	(a) What is Raspberry Pi and Li-Fi ?	03
	(b) Explain features of Arduino architecture.	04
	(c) What is CoAP and MQTT? List out the key features of CoAP.	07
OR		
Q.5	(a) What is man in the middle attack?	03
	(b) Explain application of IOT in water monitoring system.	04
	(c) What are the security and private issues of IOT? Explain using suitable case scenario.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2023****Subject Code:3160716****Date:10-07-2023****Subject Name:IOT and applications****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) List out enabling technologies in IOT. Explain any one in details	03
	(b) Explain characteristics of the IoT.	04
	(c) Draw and Explain ATMEGA328P Microcontroller Pin diagram.	07
Q.2	(a) Explain MQTT Broker and client.	03
	(b) Differentiate MQTT and CoAP.	04
	(c) Explain IoT Technology Stack.	07
	OR	
	(c) Draw and explain IPv6 Header.	07
Q.3	(a) Explain IOT messaging protocol in detail.	03
	(b) Difference between Microcontroller and Microprocessor.	04
	(c) What is ARM? Explain special feature of ARM processor.	07
	OR	
Q.3	(a) Explain Ultrasonic sound sensor.	03
	(b) Write Raspberry Pi code to blink LED ON and OFF,	04
	(c) What is BLE? Explain components of BLE.	07
Q.4	(a) Explain Various application of IoT in Food and Healthcare.	03
	(b) Explain CoAP. List the key features of CoAP	04
	(c) What is fog computing? List the characteristics of fog computing.	07
	OR	
Q.4	(a) Explain Various application of IoT in Retail.	03
	(b) Explain limitation of cloud computing.	04
	(c) Discuss various types of Cloud Computing in IOT.	07
Q.5	(a) Explain IoT and cyber physical system	03
	(b) Describe IOT security needs and issues.	04
	(c) What is Arduino? Explain features of Arduino architecture.	07
	OR	
Q.5	(a) What are the challenges of IOT security?	03
	(b) Explain any one algorithm for IOT security	04
	(c) Explain various security issues and need for developing IOT based application.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI(NEW) EXAMINATION – WINTER 2022****Subject Code:3160716****Date:15-12-2022****Subject Name:IOT and applications****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- | | | |
|------------|--|-----------|
| Q.1 | (a) Define IOT. Discuss advantages and disadvantages of IOT. | 03 |
| | (b) Explain different characteristics of IOT. | 04 |
| | (c) Draw and explain IOT architecture. | 07 |
| Q.2 | (a) What is M2M and IOT? | 03 |
| | (b) Explain sensor and microcontroller. | 04 |
| | (c) Discuss levels of IOT. | 07 |
| | OR | |
| | (c) Discuss various applications of IOT. | 07 |
| Q.3 | (a) What is and IOT messaging protocol? | 03 |
| | (b) Give difference between MQTT and COAP protocols. | 04 |
| | (c) List out types of IOT protocols and explain any one in brief. | 07 |
| | OR | |
| Q.3 | (a) Write short note on: Zigbee. | 03 |
| | (b) Give difference between IPV4 and IPV6 transport protocol. | 04 |
| | (c) What is BLE? Explain components of BLE. | 07 |
| Q.4 | (a) Explain features of Arduino. | 03 |
| | (b) Explain IOT challenges. | 04 |
| | (c) Draw and explain Arduino architecture. | 07 |
| | OR | |
| Q.4 | (a) Explain cloud for IOT. | 03 |
| | (b) Explain IOT for the retailing industry. | 04 |
| | (c) Explain interfacing of LED with Raspberry Pi. | 07 |
| Q.5 | (a) Explain web of things. | 03 |
| | (b) Describe IOT security needs and issues. | 04 |
| | (c) Explain security architecture in detail with diagram. | 07 |
| | OR | |
| Q.5 | (a) What are the challenges of IOT security? | 03 |
| | (b) Explain Raspberry Pi interface. | 04 |
| | (c) Draw the block diagram of Raspberry Pi and explain components of Raspberry Pi in brief. | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2022****Subject Code:3160716****Date:06/06/2022****Subject Name:IOT and applications****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) List any five IoT applications. Explain any one in details.	03
	(b) What is Sensor ? List different types of Sensors used to develop IOT	04
	(c) Draw and Explain ATMEGA328P Microcontroller Pin diagram.	07
Q.2	(a) Explain MQTT Broker and client.	03
	(b) Differentiate CoAP and MQTT.	04
	(c) Draw and explain IPv4 Header.	07
	OR	
	(c) Draw and explain IPv6 Header.	07
Q.3	(a) Explain pinMode(), digitalRead() and digitalWrite () functions of Arduino	03
	(b) Write Raspberry Pi code to change the brightness of LED.	04
	(c) Write arduino code to identify PH value of Soap , Tap water and Lemon juice using PH sensor.	07
	OR	
Q.3	(a) What is PWM? Explain usage of PWM pins with Example.	03
	(b) Write Raspberry Pi code to blink LED ON and OFF .	04
	(c) Write arduino code to transmit “Hello World – Code to demonstrate BT Communication” string on serial monitor using Bluetooth	07
Q.4	(a) Explain Various application of IoT in Food and Healthcare.	03
	(b) Which are the challenges in IoT with Cloud Computing? Explain any one.	04
	(c) List and explain various types of Cloud Deployment Models, also discuss their merits and demerits	07
	OR	
Q.4	(a) Explain Various application of IoT in Retail.	03
	(b) Explain Security in cloud with respect to IOT based application.	04
	(c) Discuss various types of Cloud Computing	07
Q.5	(a) Explain IoT and cyber physical system.	03
	(b) Explain any one algorithm for IOT security.	04
	(c) Draw and explain architecture of IOT security	07
	OR	
Q.5	(a) Explain IOT and WSN.	03
	(b) Explain IOT security challenges.	04
	(c) Explain various security issues and need for developing IOT based application.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3160716****Date:30/11/2021****Subject Name:IOT and applications****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Define following terms: IoT, Sensor, WSN. **03**
(b) Explain characteristics of the IoT. **04**
(c) Explain various levels of IoT. **07**
- Q.2** (a) List down components of IoT system. **03**
(b) Explain CSPP in ARM. **04**
(c) Explain IoT Technology Stack. **07**
- OR**
- (c) Explain challenges of IoT. **07**
- Q.3** (a) What is IP addressing? **03**
(b) Difference between Microcontroller and Microprocessor. **04**
(c) What is ARM? Explain special feature of ARM processor. **07**
- OR**
- Q.3** (a) What is heartbeat sensor? **03**
(b) Explain specification of sensor. **04**
(c) Explain with diagram IPv4 header format. **07**
- Q.4** (a) What is MQTT? **03**
(b) Explain CoAP. List the key features of CoAP **04**
(c) What is fog computing? List the characteristics of fog computing. **07**
- OR**
- Q.4** (a) List and explain cloud components. **03**
(b) Explain limitation of cloud computing. **04**
(c) What are the security challenges of IoT? **07**
- Q.5** (a) Explain in brief future factory concepts. **03**
(b) What is smart city? What are the features of IoT based smart city? **04**
(c) What is Arduino? Explain features of Arduino architecture. **07**
- OR**
- Q.5** (a) What is Raspberry Pi? **03**
(b) Explain security architecture. **04**
(c) What risks do insecure IoT devices bring to privacy and security? **07**
